

Hostel Maintenance & Issue Ticketing System

Problem Statement #07 | Campus & Academic Operations

SRN: PES1UG24CS375

Class / Section: SE Section G

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Target Actors: Hostel Resident, Maintenance Warden, Maintenance Staff

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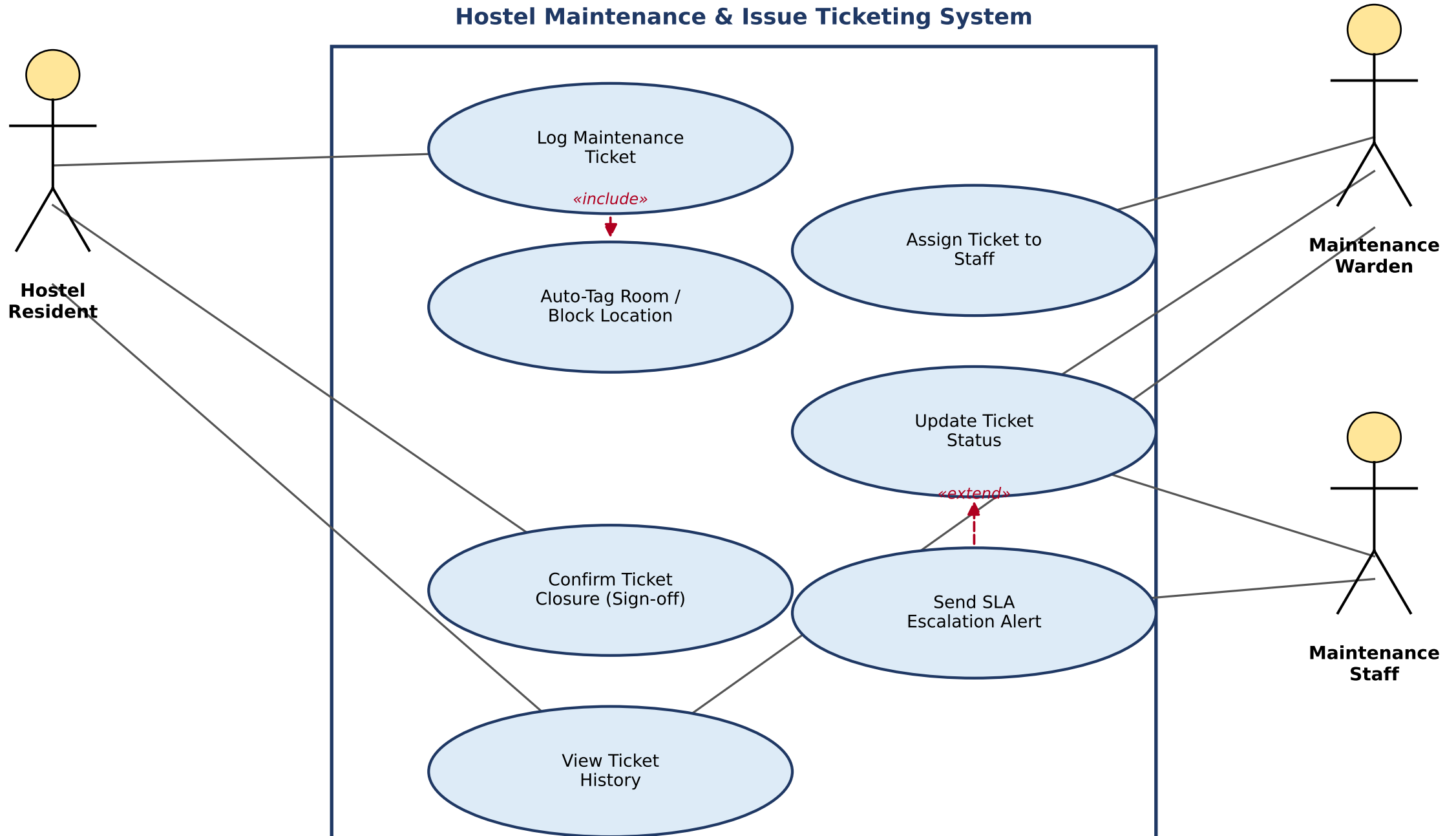
Hostel Maintenance & Issue Ticketing System

Requirements Table — SRN: PES1UG24CS375 | Problem Statement #07 | Campus & Academic Operations

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: Ticket is created, assigned a unique ID, and routed to the warden queue. Fail: An incomplete form is accepted without a room tag.	Establishes the core entry point for issue reporting with enough detail to triage.
FR-002	Functional	The system shall allow the Maintenance Warden to assign an open ticket to a Maintenance Staff member based on issue category (plumbing, electrical, carpentry) and staff availability.	High	Pass: Ticket status changes to “Assigned” and the assigned staff member is notified within 1 minute. Fail: A ticket is assigned to staff outside the matching category without an explicit override.	Ensures the correct specialist is dispatched, reducing resolution time.
FR-003	Functional	The system shall allow Hostel Residents to view the real-time status of their submitted tickets (Open, Assigned, In Progress, Resolved) from a personal dashboard.	Medium	Pass: Status shown on the dashboard reflects the warden's latest update within 2 minutes. Fail: The dashboard shows a stale or incorrect status after an update.	Improves transparency and reduces repeat status-check requests to the hostel office.
FR-004	Functional	The system shall require the Hostel Resident to digitally confirm (“sign-off”) ticket closure before a resolved ticket can be marked as fully closed.	High	Pass: A ticket cannot transition to “Closed” without a resident sign-off timestamp. Fail: A ticket is closed without resident confirmation.	Confirms the issue was actually resolved to the resident's satisfaction and avoids disputes.
FR-005	Functional	The system shall allow Maintenance Staff to log remarks, time spent, and parts used whenever they update a ticket's status.	Medium	Pass: A status update submitted without a remark is rejected with a validation prompt. Fail: A ticket moves to “Resolved” with no remarks recorded.	Creates an audit trail for maintenance work and supports SLA and resource reporting.
NFR-001	Nonfunctional	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without a status change.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Guarantees urgent issues (e.g., electrical faults) are not left unattended past a safe threshold.
NFR-002	Nonfunctional	The system shall restrict access to ticket photo attachments and resident contact details to the assigned Maintenance Warden and Staff only, via role-based access control.	High	Pass: An access-control test confirms unauthorized roles receive a denied response when requesting another resident's ticket data. Fail: Any authenticated user can view another resident's ticket photos or contact information.	Protects resident privacy given the sensitive personal data (room number, photos) being collected.

Rows shaded blue (FR-001, NFR-001) reproduce the instructor-supplied samples for reference; FR-002–FR-005 and NFR-002 are original.

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— Association
- - - «include» / «extend»

Use-Case Flow Specification

Hostel Maintenance & Issue Ticketing System — PES1UG24CS375

Use Case	Log Maintenance Ticket (UC-01)
Primary Actor	Hostel Resident
Supporting Actors	System (auto room/block tagging), Maintenance Warden (receives routed ticket)
Preconditions	The resident is authenticated and has a valid, registered room/block assignment in the system.
Postconditions	A ticket exists with a unique Ticket ID, status = “Open”, and is visible in the Maintenance Warden's queue.

Main Success Scenario

1. Resident taps “Report New Issue” on the home screen.
2. System displays a ticket form requesting issue category (Plumbing / Electrical / Carpentry), a description, and severity.
3. System auto-tags the resident's registered room number and block («include» Auto-Tag Room/Block Location).
4. Resident optionally attaches up to 3 photos of the issue.
5. Resident selects a severity level (Low / Medium / High) and submits the ticket.
6. System validates that all mandatory fields — category, description, severity, and room tag — are present.
7. System generates a unique Ticket ID and sets the ticket status to “Open.”
8. System routes the ticket to the Maintenance Warden's queue and sends a confirmation notification to the resident.
9. Use case ends successfully.

Alternate Flow — 6a. Missing Mandatory Field

- 6a1. System detects that the room tag or category field is missing.
- 6a2. System displays an inline validation error highlighting the missing field(s).
- 6a3. Resident corrects the field(s) and resubmits, returning to step 6.